

Assignment: 02

1. The Metadata Packing System

Scenario: When an AI agent processes a user request, it needs to generate a quick "metadata packet" containing the task type, the priority level, and the timestamp.

- **The Task:** Create three variables holding a task name, a priority, and a year. Pack them into an immutable **Tuple**. Then, access and print just the priority level from that tuple.
- **Sample Input:** task = "Data Cleanup", priority = "High", year = 2026
- **Sample Output:** "Extracted Priority: High"

2. The AI Context Truncator

Scenario: LLMs have limits on how much text they can read. If a user inputs a long paragraph, you often need to grab just the very beginning and the very end to create a shorthand snippet.

- **The Task:** Take a long string. Use **String Slicing** to extract the first 10 characters and the last 10 characters, then combine them with ... in the middle.
- **Sample Input:** "Agentic systems are completely rewriting software."
- **Sample Output:** "Agentic sy...e software."

3. The Reverse Token Encryption (String Slicing with Step)

Scenario: To keep sensitive API data secure during transmission, an agent runs a simple security obfuscation step: it reverses a secret key string and extracts every second character.

- **The Task:** Take a 12-character string variable. Use **String Slicing with a negative step** to reverse the entire string while skipping every other character.
- **Sample Input:** "AGENT_SECRET"
- **Sample Output:** "TERES_A"

4. The Agent Memory Stack

Scenario: An agent needs to track the history of an active conversation. As the conversation progresses, new topics are added to the list, and old ones might be cleared out.

- **The Task:** Create a **List** with 3 starter topics. Add a 4th topic to the end of the list. Then, remove the very first topic because it's too old. Finally, print the total number of topics left in memory.
- **Sample Initial Data:** ["Weather", "Math", "Coding"]
- **Sample Output:** ["Math", "Coding", "Deep Learning"] | Total: 3

5. The Memory Buffer Organizer (Advanced List Methods)

Scenario: An AI agent logs the performance scores of its previous runs. Before analyzing the data, you need to clear out errors, rank the remaining scores, and find the absolute best and worst performances.

- **The Task:**
 1. Create a list of scores that includes an accidental string error: [88, 92, "ERROR", 74, 92, 85].
 2. Find the index of the "ERROR" string and **remove** it.
 3. Find out how many times the score 92 appears using **count**.
 4. **Insert** a missing score of 95 at the second position (index 1).
 5. **Sort** the list in ascending order.
 6. Find and print the **min** (lowest) and **max** (highest) scores in the buffer.
- **Sample Output:** Final List: [74, 85, 88, 92, 92, 95] | Min: 74 | Max: 95 | Count of 92: 2